Week 2 Problem Set

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```
#####################################################
## title: second problem set!
## author: you!
## purpose: to try out R
## date: today's date
#####################################################

# you can start coding below
```

1. On GitHub, create a new **repository** called "soc10problemsets" (or something similar). Then upload the "Week2PSetTemplate.Rmd" file (you can download it here here). Make Tyler (`tylermcdaniel`) and Yao (`yao-mxu`) "collaborators" in this repository so we can see your work (Settings -> Collaborators -> Add People). Add your name and the date to the top of the file, save your changes, and **commit** them. (note: You can do this online, but I recommend using GitHub Desktop!).

Here I created a new repository called soc10problemsets and shared that with both Tyler and Yao!

2. In the space below, set "eval = TRUE" and try running the first code chunk. Then fix the error. On your local file, **commit** the file to your GitHub.

For question 2 I found three errors. First, I had to change the eval = FALSE to evaul = TRUE. Next, I had to change the error within new_dataframe -> data.frame and fix the arrow from facing right to facing left new_dataframe <- data.frame. Lastly, I loaded the package library(dplyr).

```
# this is an initial dataframe
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
df1 <- data.frame("resource name" = "a",
                  "custom_ready" = "b")


# we want to add observations to it (c, and d)
# something is wrong with the following function - can you fix it?
add_line <- function(dataframe, new_dataframe){

  new_dataframe <- data.frame("resource" = "c",
                              "custom_ready" = "d")

  bind_rows(dataframe, new_dataframe)

}

# this should return a dataframe with all our observations (a, b, c, and d, neatly organized)
add_line(df1)
```

```
##   resource.name custom_ready resource
## 1             a            b     <NA>
## 2          <NA>            d        c
```

3. Add your own data to a collaboration! In the markdown file, add to the small dataframe with some potential data sources for further social science research. Add one or more resources to this list. **Commit** your changes.

```r
resources_df <- data.frame("resource name" = c("All Our Ideas", "Education and Income (Census)"),

"link" = c("https://allourideas.org/", "https://www.census.gov/library/stories/2025/09/education-and-in

"custom_ready" = c("custom made", "not custom"),

"description" = c("Education and Income is a U.S Census Bureau story that
                  shows how income varies by educaitonal level. It demonstrates
                  that higher levels of education are associated with higher
                  earnings and differences in income outcomes across the population.")
)
```

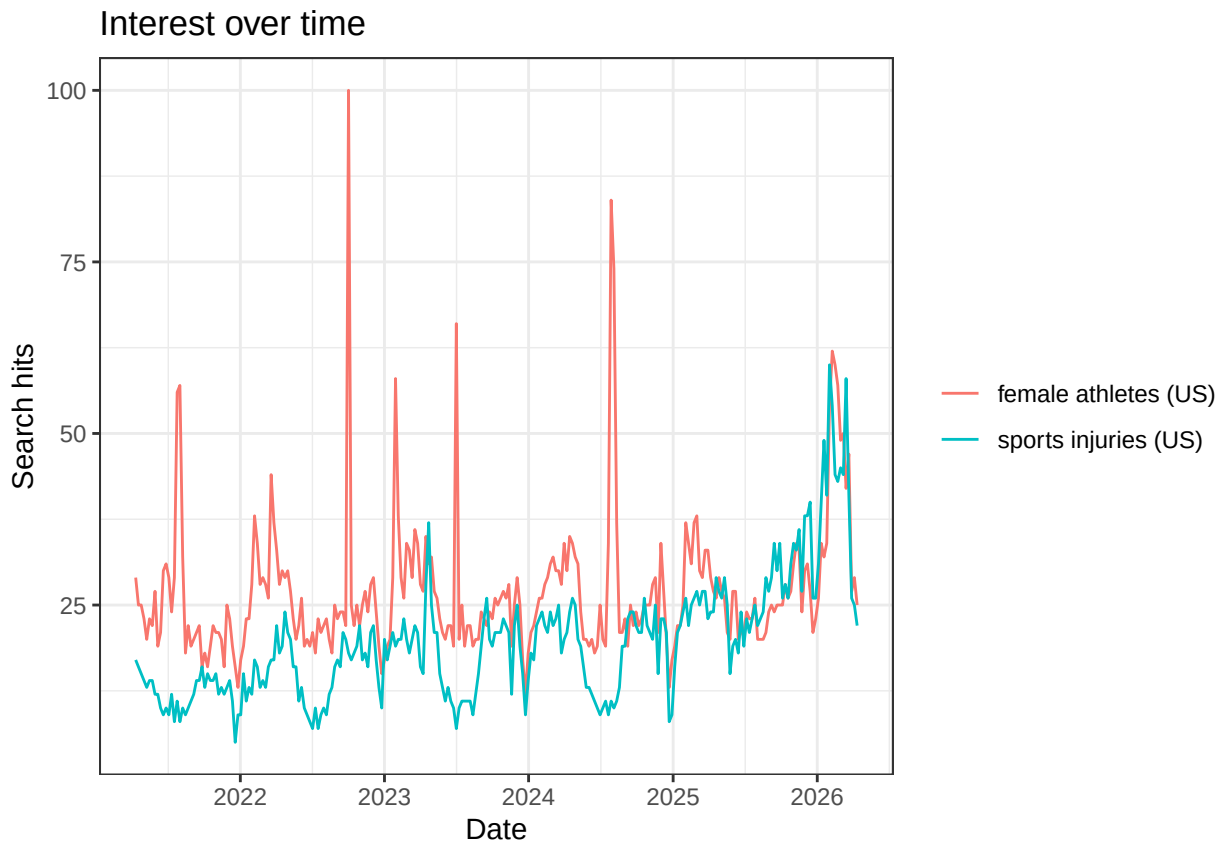
4. Using the `gtrendsR` R package, download data on a term of your choosing. Show a plot of the search term over time, and then plot it over different geographies. Discuss how each of Salganik's 10 characteristics of computational data apply (or don't apply). **Commit** your changes.

I downloaded the gtrendsR package, and chose to download data on female athletes and sports injuries as I am a college athlete I thought it could be interesting. When considering Salganik's 10 characteristics of computational data this is how I think it does or does not apply. 1. Always on: this applies because the trend data is continuously collecting information as people search, so this dataset represents a report over a long period time and ongoing. 2. Nonreactive: This also applies because users do not know they are being used for a research study, making this collection natural, and not biased by people when they search. 3. Incomplete: This applies because not everyone uses a search for things regarding this topic, and so only use the information given within gtrendsR but it is not all data it is just a sample. 4. Inaccessible: This applies because Google does not have complete access to raw Google search data, and only get to see normalized trends. 5 Nonrepresentative: This applies as users of Google do not represent the full population but search

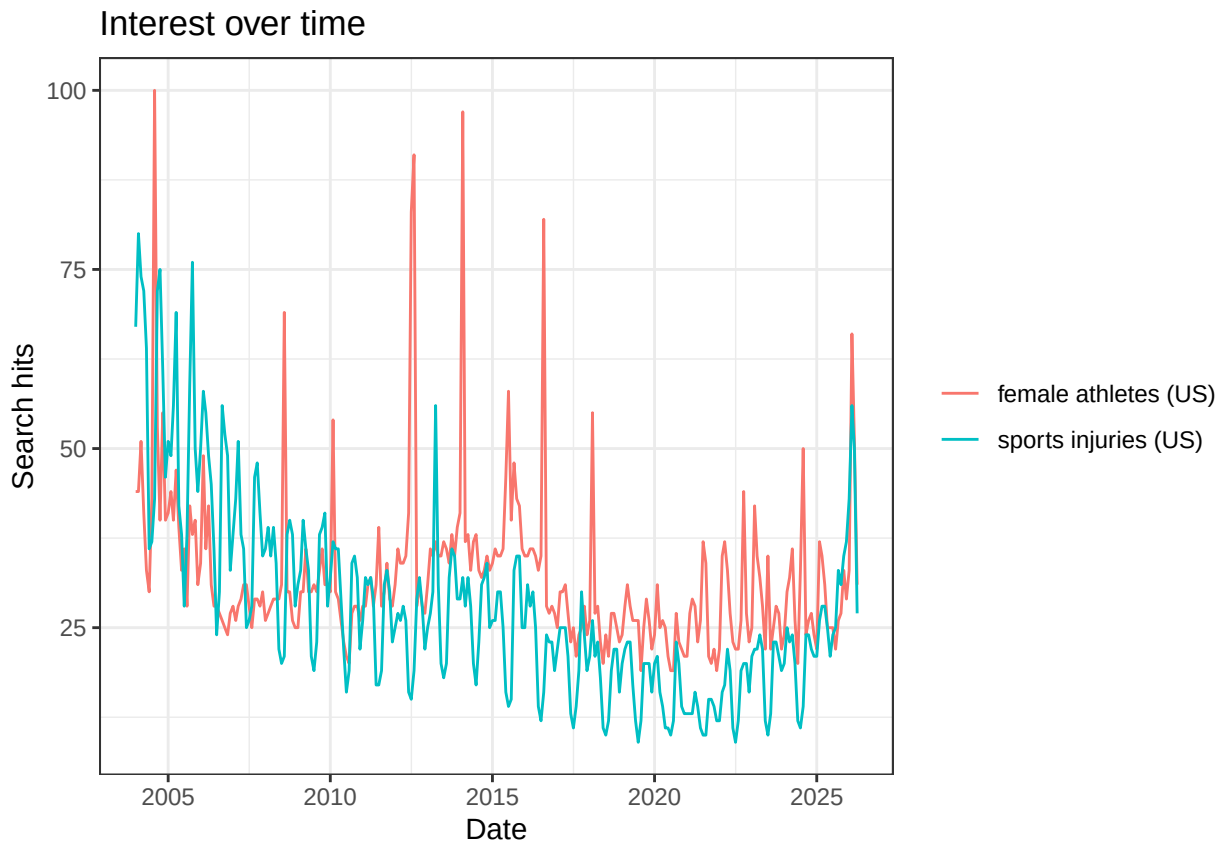
users, which biases the data and could more heavily be athletes, or coaches. 6. Drifting: This applies because the data evolves and changes over time, as female sports and information on injuries are becoming more popular then they were a long time ago and so in turn might have increased searches. For example, ACL's being recently seen as very common in female athletes. 7. Algorithmically confounded: This applies because Google's algorithm may impact search by auto suggesting topics, or ads that pop up that may influence what a user searches. 8. Dirty: This applies because the data requires cleaning before it can be used and analyzed. 9. Sensitive: This is anonymous so does not impact the data, or have direct sensitivity. 10. Distributed: This applies because access depends on Google not the researcher having the raw data. In my personal example that I chose most of Salganik's charachteristics apply to gtrendsR and show it's strengths like being always on and nonreactive, while also showing limitations like being incomplete, and nonrepresentiative.

```r
library(gtrendsR)

data_social <- gtrends(c("female athletes", "sports injuries"),
  geo = c("US")
)

plot(data_social)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the gtrendsR package.
##   Please report the issue at <https://github.com/PMassicotte/gtrendsR/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
data_social_all <- gtrends(c("female athletes", "sports injuries"),
  geo = c("US"), time = "all"
)

plot(data_social_all)
```




```
data_geo <- gtrends(c("female athletes"),
  geo = c("US-CA", "US-NY", "US-FL"), time = "all"
)
```

5. List the three types of collaboration that Salganik names. When can each work well, and what are potential downsides? List one new idea for a collaborative data project in each category. **Commit** your changes.

The three types of collaboration that Salganik names are human computation, open calls, and distributed data collection. Human computation works for tasks that are easy for humans because they are better for recognition and labeling tasks and harder for computers. However, a downside is that it can require a lot of participants and can take a long time. A new idea for this is volunteers logging on to classify ocean data, such as identifying marine species and maybe tagging underwater images to help environmental research. Open calls works well for when a problem can be casted wide and many people with the skills can contribute and try to find possible solutions. A downside is that may be limited to those who have the skills to solve or contribute. A new idea for this is a competition where participants improve recommendation algorithms for a streaming platform like Hulu to help better match users with content associated with their viewing preferences. Distributed data collection works well when a large amount of participation is needed to collect real world data across an immense amount of locations. A downside is that the quality of the data can depend on the participants, and lack full and equal representations of all locations. A new idea for this is something like In the Wild, a platform that allows users to report plant observations and species data to help researchers better understand plants growth, distribution, and conservation.